

MARIANA YEPES

Data Scientist | Mathematical Engineer | ML & Predictive Modeling

marianayepes98@hotmail.com • +57 (300) 873-8052 • Medellín, Colombia • Remote Ready

TECHNICAL SKILLS

ML & Modeling:	Regression, classification, clustering, route optimization, heuristic algorithms, simulation models, predictive modeling
Deep Learning:	TensorFlow, PyTorch — neural network architectures, model training & evaluation
Python Stack:	Scikit-learn, Pandas, NumPy, Flask, Pydeck, GeoJSON, Jupyter Notebooks
Data & BI:	SQL, R Studio, Power BI, Tableau, geospatial analysis, data wrangling
Tools:	Git/GitHub, Notion, Microsoft Office
Languages:	Spanish (Native), English (C1 – Advanced), Italian (Fluent), French (Intermediate), Portuguese (B2)

PROFESSIONAL SUMMARY

Data Scientist and Mathematical Engineer with 3+ years of hands-on experience building ML models — regression, classification, clustering, optimization, and simulation — applied to real-world operational problems. Research background includes deep learning (TensorFlow, PyTorch) and advanced quantitative modeling, with a peer-reviewed publication in an Elsevier Q1 journal. Strong ability to translate complex data problems into actionable solutions and communicate findings clearly to both technical and business stakeholders. Bilingual (English C1 / Spanish Native), collaborative, and eager to bring rigorous analytical thinking to a high-impact telecom environment.

PROFESSIONAL EXPERIENCE

Business & Product Analyst / Product Manager | **Tributi – Fintech Tax Automation SaaS** | **Colombia (Remote)** *Sept 2024 – Present*

- Analyzed operational data each tax cycle to identify process inefficiencies and model improvements, translating findings into actionable recommendations for product and engineering teams.
- Managed data quality and consistency across multiple sources, ensuring reliable reporting and decision-making throughout each delivery cycle.
- Applied data-driven resource allocation to manage OCR certificate processing budget, optimizing cost-efficiency across annual processing volumes.

Financial Data Analyst | **Marsan Foods** | **Canada (Remote)** *May 2024 – Sep 2024*

- Analyzed pricing datasets and built BI dashboards (Power BI, Tableau) tracking KPIs, enabling data-driven decisions for senior stakeholders across business reviews.

Lead Researcher – ML & Predictive Modeling | **EAFIT University – GRID Research Group** | **Medellín, Colombia** *Jan 2023 – Jan 2024*

- Built regression, classification, and clustering models to analyze logistics and operational data for real-world business clients, identifying patterns that reduced operational costs and improved decision-making.
- Developed route optimization algorithms and heuristic models for last-mile logistics, applying combinatorial optimization techniques to solve NP-hard routing problems.

- Designed simulation models to evaluate operational scenarios and predict performance under varying conditions, supporting strategic planning for industry clients.
- Translated complex business requirements into mathematical and technical specifications, bridging the gap between domain experts and the modeling team.
- **Research output:** Co-authored peer-reviewed publication:
- "Charging policies for battery swapping stations for hybrid motorcycles" — Cleaner Logistics and Supply Chain, Elsevier Q1 (2025). Presented findings at IFORS 2023 International Conference, Chile.

Economy Department Researcher | **EAFIT University | Medellín, Colombia** *May 2022 – Dec 2022*

- Delivered quantitative analytical insights for a University of Illinois-funded research project, managing structured reporting to cross-institutional stakeholders within scope and schedule.

Operations & Logistics Researcher – Python Modeling | **EAFIT University – GRID Research Group | Medellín, Colombia** *Sep 2021 – Sep 2022*

- Developed an optimal route viewer for courier logistics using Python (Flask, Pydeck, GeoJSON), applying process modeling and geospatial data analysis to support real-time operational decisions.
- Built data pipelines to process and visualize route data, enabling stakeholders to evaluate optimization outcomes interactively.

EDUCATION

B.S. Mathematical Engineering — EAFIT University & Concordia University (Montreal, Canada)

B.A. Social Communication and Journalism — EAFIT University